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Project Report

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Project details

Title of the project	**AI-Based Road Damage and Pothole Detection**

Table Of Contents

1. Introduction.....3
2. Objectives..... 3
3. Methodology and Implementation.....4
4. Results.....4
5. Conclusion..... 4

1. Introduction

Roads form the backbone of a country's transportation infrastructure, and their condition directly affects travel safety, vehicle maintenance costs, and daily commute efficiency. Over time, roads develop various forms of damage such as cracks and potholes due to heavy traffic load, weathering, poor drainage, and ageing of construction material. In India and many other countries, road damage is one of the leading contributors to accidents, vehicle breakdowns, and traffic congestion, yet its inspection and reporting is still largely a manual process.

Traditionally, road surveys are carried out by government authorities or municipal bodies through manual visual inspection, where trained personnel physically travel along road stretches and note down damaged sections. This method is slow, labour-intensive, expensive, and subjective — different inspectors may rate the same damage differently, and large road networks (especially rural and semi-urban ones) often go unsurveyed for months or years due to resource constraints.

With the rapid advancement of Artificial Intelligence and Computer Vision, it is now possible to automate this inspection process using image and video data. This project, "AI Based Road Damage and Pothole Detection," proposes a deep learning-based system that can automatically detect and classify road surface damage from images or video captured using a standard smartphone camera. The system focuses on identifying and categorizing damage into four major classes — Longitudinal Cracks, Transverse Cracks, Alligator Cracks, and Potholes — which are the most common and internationally recognized forms of flexible pavement distress. By automating detection and classification, this project aims to make road condition monitoring faster, cheaper, more consistent, and scalable to large road networks.

2. Objectives

The following are the Objectives We will meet in our project.

- To develop an AI-based system capable of automatically detecting road surface damage from images/video captured using a standard camera.
- To classify detected damage into four distinct categories: Longitudinal Crack, Transverse Crack, Alligator Crack, and Pothole, using a supervised deep learning model.
- To train and evaluate a Convolutional Neural Network (CNN) / object detection model ( ) on a labelled road damage dataset for high detection accuracy. To build a preprocessing pipeline (resizing, noise removal, augmentation) that improves model robustness against varying lighting, road surface, and camera angle conditions.

- To generate a structured report/dashboard output showing detected damage type, location (via GPS tagging where available), and severity for easier maintenance planning.
- To evaluate the model's performance using standard metrics such as precision, recall, F1-score, and mean Average Precision (mAP).

3. Methodology and Implementation

4. Results

5. Conclusion